

Learning Activity 10: Raking for Nonresponse Adjustment

STA304

Goal

Raking is a weighting adjustment method used when we know population margins for two or more variables, but we do not know the full population counts for every cell in the post-stratification.

The outcome of interest is the total number of weekly study hours. You suspect that women are less likely to answer this than men question and that the year of the study will also affect response propensities. You worry because you suspect both of these factors (sex and year of study) are also correlated with the total number of weekly study hours.

The number of females in the population is 60, the number of males in the target population is 40. The number of first years, second years, and third years in the target population are 25, 35, and 40. There are 100 people in the population.

Part 1: Sample the population and calculate weights for the sample

Suppose an SRS of ($n=50$) students was selected from a population of ($N=100$) students.

```
set.seed(2)
srs_sample <- population[sample(1:100, 50, replace = FALSE),]

srs_sample %>% ungroup() %>% select(sex, year, obs_hours) %>% head(10)
```

```
##      sex year obs_hours
## 85  Male  3rd      14.4
## 79  Male  3rd      13.8
## 70  Male  2nd      15.5
## 6   Male  2nd      13.2
## 32  Male  2nd      14.8
## 8   Female 3rd       NA
## 17 Female 3rd       NA
## 93  Male  1st      15.3
## 81 Female 3rd       NA
## 76 Female 3rd       NA
```

Among the respondents, the observed respondent counts are:

```
srs_sample %>% ungroup() %>% select(sex, year, obs_hours) %>%
  filter(!is.na(obs_hours)) %>%
  group_by(sex, year) %>%
  summarise(n_obs = length(obs_hours))
```

```
## # A tibble: 6 x 3
## # Groups:   sex [2]
##   sex    year n_obs
##   <fct> <fct> <int>
## 1 Female 1st     6
## 2 Female 2nd     2
## 3 Female 3rd     4
## 4 Male   1st     6
## 5 Male   2nd     5
## 6 Male   3rd     6
```

a) What is the initial survey weight for each respondent?

b) Create an original weight table for the respondents as shown in the slides on page 47

	Year 1	Year 2	Year 3	Sum of weights
Female				
Male				
Sum of weights				

Part 2: Why do we need raking?

a) Does the initial respondent table match the known gender margins? What about the known year margins?

b) Why can we not simply post-stratify?

Part 3: First row adjustment

First, adjust the rows so that the row totals match the known gender totals.

The row adjustment factor is

$$\text{row adjustment factor} = \frac{\text{target row total}}{\text{current row total}}.$$

- a) Compute the row adjustment factor for Female.
- b) Compute the row adjustment factor for Male.
- c) Fill in the adjusted table after applying the row adjustment factors.

	Year 1	Year 2	Year 3	Row total
Female				60
Male				40
Column total				100

- d) After this row adjustment, do the column totals match the known year totals?

Part 4: First column adjustment

Now adjust the columns so that the column totals match the known year totals.

The column adjustment factor is

$$\text{column adjustment factor} = \frac{\text{target column total}}{\text{current column total}}.$$

Use your column totals from Part 2.

- a) Compute the column adjustment factor for Year 1.

b) Compute the column adjustment factor for Year 2.

c) Compute the column adjustment factor for Year 3.

d) Fill in the adjusted table after applying the column adjustment factors.

	Year 1	Year 2	Year 3	Row total
Female				
Male				
Column total	25	35	40	100

e) After this column adjustment, do the row totals still match the known gender totals?

Part 5: Second row adjustment

Repeat the row adjustment using the table from Part 3.

a) Compute the new row adjustment factor for Female.

b) Compute the new row adjustment factor for Male.

c) Fill in the adjusted table after applying this second row adjustment.

	Year 1	Year 2	Year 3	Row total
Female				60
Male				40
Column total				100

Part 6: Raking until convergence in R

The following code alternates between row adjustments and column adjustments until the row and column totals are close enough to the population margins.

Read the code and then answer the discussion questions below.

```
# Current weighted respondent table
w <- matrix(c(12, 4, 8,
              12, 10, 12),
            nrow = 2, byrow = TRUE)

rownames(w) <- c("Female", "Male")
colnames(w) <- c("Year 1", "Year 2", "Year 3")

# Known population margins
row_targets <- c(Female = 60, Male = 40)
col_targets <- c("Year 1" = 25, "Year 2" = 35, "Year 3" = 40)

# Settings
max_iter <- 100
tol <- 1e-6

# Store a copy of the starting table
raked <- w

for (iter in 1:max_iter) {

  # Step 1: adjust rows
  row_factors <- row_targets / rowSums(raked)
  raked <- sweep(raked, 1, row_factors, FUN = "*")

  # Step 2: adjust columns
  col_factors <- col_targets / colSums(raked)
  raked <- sweep(raked, 2, col_factors, FUN = "*")

  # Check how far the current margins are from the targets
  row_error <- max(abs(rowSums(raked) - row_targets))
  col_error <- max(abs(colSums(raked) - col_targets))
  max_error <- max(row_error, col_error)

  cat("Iteration:", iter, "Maximum margin error:", max_error, "\n")

  # Stop once the margins are close enough
  if (max_error < tol) {
    break
  }
}
```

```
## Iteration: 1 Maximum margin error: 3.470643
```

```
## Iteration: 2 Maximum margin error: 0.1019516
## Iteration: 3 Maximum margin error: 0.002992803
## Iteration: 4 Maximum margin error: 8.785217e-05
## Iteration: 5 Maximum margin error: 2.578853e-06
## Iteration: 6 Maximum margin error: 7.570083e-08
```

```
raked
```

```
##           Year 1   Year 2   Year 3
## Female 17.778552 17.36566 24.85579
## Male    7.221448 17.63434 15.14421
```

```
rowSums(raked)
```

```
## Female   Male
##      60    40
```

```
colSums(raked)
```

```
## Year 1 Year 2 Year 3
##     25    35    40
```

Part 7: Discussion questions

- a) What does the sweep function do in this context?

- b) What does it mean for the algorithm to converge?

- c) In the code, what role does the tolerance value play?

Part 8: Estimating a mean after raking

The mean for the observed data is:

```
summary_table <- srs_sample %>% ungroup() %>% select(sex, year, obs_hours) %>%  
  filter(!is.na(obs_hours)) %>%  
  group_by(sex, year) %>%  
  summarise(n_obs = length(obs_hours),  
            mean_hours = mean(obs_hours))
```

summary_table

```
## # A tibble: 6 x 4  
## # Groups:   sex [2]  
##   sex    year n_obs mean_hours  
##   <fct> <fct> <int>      <dbl>  
## 1 Female 1st     6       13.2  
## 2 Female 2nd     2       17.9  
## 3 Female 3rd     4       23.0  
## 4 Male   1st     6        9.15  
## 5 Male   2nd     5       14.9  
## 6 Male   3rd     6       19.4
```

- a) Use the final raked weights from the R code to estimate the total number of weekly study hours. What would have been the estimate with ignoring nonresponse (SRS estimate with $n=29$, $N=100$)? The population total hours is 1715.8. Compare.

- b) Use your raked estimated total to estimate the population mean number of weekly study hours. The population mean hours is 17.158. Compare.